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MEDICAL MANIPULATOR AND MEDICAL MANIPULATOR SYSTEM

Abstract

A medical manipulator includes an outer sheath and a channel tube that is inserted into the outer sheath. At least a part of the channel tube includes a coil sheath forming a lumen into which a treatment manipulator or a treatment tool is inserted and a blade tube disposed outside of the coil sheath.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] Priority is claimed on U.S. Provisional Patent Applications (1) No. 63/562,855 filed on Mar. 8, 2024, (2) No. 63/635,021 filed on Apr. 17, 2024, (3) No. 63/648,937 filed on May 17, 2024, (4) No. 63/664,879 filed on Jun. 27, 2024, (5) No. 63/669,306 filed on Jul. 10, 2024, (6) No. 63/688,972 filed on Aug. 30, 2024, (7) No. 63/691,009 filed on Sep. 5, 2024, (8) No. 63/695,602 filed on Sep. 17, 2024, and (9) No. 63/737,339 filed on Dec. 20, 2024, the contents of which are incorporated herein by reference.

FIELD
[0002] The present invention relates to a medical manipulator and a medical manipulator system.
BACKGROUND
[0003] In the related art, a medical manipulator system for observation or treatment in a hollow organ such as the alimentary canal has been used. The medical manipulator system can electrically drive an insertion section or the like which is inserted into a hollow organ. A user can control the operation of the insertion section or the like using an operation device which is provided outside of an internal part.
[0004] In PCT International Publication No. WO 2021/145411 (hereinafter referred to as Patent Document 1), a medical system including an endoscope which is electrically driven is described. In the medical system described in Patent Document 1, since the endoscope is electrically driven, it is possible to reduce fatigue of an operator.
[0005] The medical manipulator system according to the related art described in Patent Document 1 is not normally easily used and is not a system that can efficiently perform treatment using a medical manipulator (such as an endoscope).
SUMMARY
[0006] The present invention provides a medical manipulator and a medical manipulator system that can more efficiently perform observation or treatment.
[0007] A medical manipulator according to a first aspect of the present invention includes an outer sheath and a channel tube that is inserted into the outer sheath, wherein at least a part of the channel tube includes a coil sheath forming a lumen into which a treatment manipulator or a treatment tool is inserted and a blade tube disposed outside of the coil sheath.
[0008] With the medical manipulator and the medical manipulator system according to the present invention, it is possible to more efficiently perform observation or treatment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** is a diagram illustrating the entire configuration of an electric endoscope system

according to a first embodiment.
[0010] FIG. **2** is a diagram illustrating an insertion manipulator of an electric endoscope system which is inserted into a large intestine.
[0011] FIG. **3** is a diagram illustrating a distal end portion of an insertion section of the insertion manipulator.
[0012] FIG. **4** is a front view of the distal end portion when seen from the distal end side.
[0013] FIG. **5** is a sectional view of the distal end portion.
[0014] FIG. **6** is a diagram illustrating a bendable portion of the insertion manipulator.
[0015] FIG. **7** is a diagram illustrating joint rings of the bendable portion.
[0016] FIG. **8** is a diagram illustrating a first channel tube of the insertion manipulator.
[0017] FIG. **9** is a sectional view of a proximal end channel tube of the first channel tube.
[0018] FIG. **10** is a functional block diagram of a drive device.
[0019] FIG. **11** is a functional block diagram of a video control device.
[0020] FIG. **12** is a diagram illustrating a modified example of the bendable portion.
[0021] FIG. **13** is a diagram illustrating a first channel tube which is inserted into the modified example of the bendable portion.
[0022] FIG. **14** is a diagram illustrating a modified example of the insertion section.
[0023] FIG. **15** is a diagram illustrating a spiral tube.
[0024] FIG. **16** is a diagram illustrating a modified example of the distal end portion.
[0025] FIG. **17** is a diagram illustrating a hardness changing device according to a second embodiment.
[0026] FIG. **18** is a diagram illustrating the entire configuration of the hardness changing device.
[0027] FIG. **19** is a diagram illustrating a hardness changing unit.
[0028] FIG. **20** is a diagram illustrating the hardness changing unit.
[0029] FIG. **21** is a diagram illustrating an operation of the hardness changing unit.
[0030] FIG. **22** is a diagram illustrating an operation of the hardness changing unit.
[0031] FIG. **23** is a diagram illustrating an operation of the hardness changing unit.
[0032] FIG. **24** is a diagram illustrating an operation of the hardness changing unit.
[0033] FIG. **25** is a diagram illustrating an operation of the hardness changing unit.
[0034] FIG. **26** is a diagram illustrating a modified example of the hardness changing unit.
[0035] FIG. **27** is a diagram illustrating the modified example of the hardness changing unit.
[0036] FIG. **28** is a diagram illustrating another modified example of the hardness changing unit.
[0037] FIG. **29** is a diagram illustrating the modified example of the hardness changing unit.
[0038] FIG. **30** is a diagram illustrating another modified example of the hardness changing unit.
[0039] FIG. **31** is a diagram illustrating the modified example of the hardness changing unit.
[0040] FIG. **32** is a diagram illustrating another modified example of the hardness changing unit.
[0041] FIG. **33** is a diagram illustrating a high-frequency knife in a manipulator tool according to a third embodiment.
[0042] FIG. **34** is a diagram illustrating a high-frequency knife for marking.
[0043] FIG. **35** is a diagram illustrating a local injection needle.
[0044] FIG. **36** is a diagram illustrating the local injection needle that locally injects a liquid.
[0045] FIG. **37** is a diagram illustrating the high-frequency knife for incision.
[0046] FIG. **38** is a diagram illustrating the high-frequency knife for incision.
[0047] FIG. **39** is a diagram illustrating a basket.
[0048] FIG. **40** is a diagram illustrating the basket for recovering a target part.
[0049] FIG. **41** is a diagram illustrating a manipulator tool according to a fourth embodiment.
[0050] FIG. **42** is a sectional view of the manipulator tool.
[0051] FIG. **43** is a diagram illustrating a modified example of the manipulator tool.
[0052] FIG. **44** is a diagram illustrating an artificial muscle disposed at another position.
[0053] FIG. **45** is a diagram illustrating an artificial muscle disposed at another position.

[0054] FIG. **46** is a diagram illustrating an artificial muscle disposed at another position.
[0055] FIG. **47** is a diagram illustrating an artificial muscle disposed at another position.
[0056] FIG. **48** is a diagram illustrating an artificial muscle disposed at another position.
[0057] FIG. **49** is a diagram illustrating another modified example of the manipulator tool.
[0058] FIG. **50** is a diagram illustrating a suture device according to a fifth embodiment.
[0059] FIG. **51** is a diagram illustrating a captured image captured by a scope of the insertion manipulator.
[0060] FIG. **52** is a diagram illustrating a needle that is delivered from a first jaw to a second jaw in the suture device.
[0061] FIG. **53** is a diagram illustrating the needle that is delivered from the first jaw to the second jaw in the suture device.
[0062] FIG. **54** is a diagram illustrating the needle that is delivered from the first jaw to the second jaw in the suture device.
[0063] FIG. **55** is a diagram illustrating the needle that is delivered from the first jaw to the second jaw in the suture device.
[0064] FIG. **56** is a diagram illustrating a sutured defective part.
[0065] FIG. **57** is a diagram illustrating an operation of an anchor applier.
[0066] FIG. **58** is a diagram illustrating an operation of the anchor applier.
[0067] FIG. **59** is a diagram illustrating an operation of the anchor applier.
[0068] FIG. **60** is a diagram illustrating an operation of the anchor applier.

DETAILED DESCRIPTION

First Embodiment

[0069] An electric endoscope system **1000** according to a first embodiment of the present invention will be described below with reference to FIGS. **1** to **13**. FIG. **1** is a diagram illustrating the entire configuration of the electric endoscope system **1000** according to the present embodiment. The electric endoscope system **1000** is an example of a medical manipulator system. A medical manipulator includes an insertion manipulator **100** which is inserted into a human body and an endoscope, a catheter, a treatment tool, and an endoluminal device which are electrically driven.

[Electric Endoscope System **1000**]

[0070] The electric endoscope system **1000** is a medical system that observes and treats an internal part of a patient. The electric endoscope system **1000** includes an insertion manipulator **100**, a treatment manipulator **400**, a drive device **500**, a video control device **600**, an operation device **800**, and a display device **900**.

[0071] FIG. **2** is a diagram illustrating an insertion manipulator **100** which is inserted into a large intestine.

[0072] The insertion manipulator **100** is a device that is inserted into a lumen of a patient to observe and treat a lesion. The insertion manipulator **100** has excellent insertability and can be inserted to, for example, an ascending colon AC or a cecum CE of a large intestine as illustrated in FIG. **2**. The insertion manipulator **100** is detachably attached to the drive device **500** and the video control device **600**. An internal path **101** is formed in the insertion manipulator **100**. In the following description, a side of the insertion manipulator **100** which is inserted into a lumen of a patient is referred to as a “distal end side (distal side) A**1**,” and a side which is attached to the drive device **500** is referred to as a “proximal side (proximal side) A**2**.”

[0073] The treatment manipulator **400** is, for example, a device that is inserted into a first channel tube **171** of the insertion manipulator **100**, protrudes from a first opening **111***a*, and is inserted into a lumen of a patient to treat a lesion. An end effector (a treatment portion) that treats a lesion is disposed at a distal end of the treatment manipulator **400**.

[0074] The drive device **500** is detachably connected to the insertion manipulator **100** and the operation device **800**. The drive device **500** drives a motor built thereinto on the basis of an operation input to the operation device **800** to electrically drive the insertion manipulator **100**. The drive device **500** drives a pump or the like built thereinto on the basis of an operation input to the operation device **800** and causes the insertion manipulator **100** to perform air supply, water supply, and suction.

[0075] The video control device **600** is detachably connected to the insertion manipulator **100** and acquires a captured image from the insertion manipulator **100**. The video control device **600** causes the display device **900** to display the captured image acquired from the insertion manipulator **100** or a GUI image or a CG image for providing information to an operator.

[0076] The drive device **500** and the video control device **600** constitute a control device **700** that controls the electric endoscope system **1000**. The control device **700** may further include peripherals such as a video printer. The drive device **500** and the video control device **600** may be a unified device.

[0077] The operation device **800** is detachably connected to the drive device **500** via an operation cable **801**. The operation device **800** may communicate with the drive device **500** in a wireless communication manner instead of in a wired communication manner. An operator S can electrically drive the insertion manipulator **100** by operating the operation device **800**.

[0078] The display device **900** is a device that can display an image such as an LCD. The display device **900** is connected to the video control device **600** via a display cable **901**.

[0079] The operation device **800** and the display device **900** are mounted in a trolley. The trolley on which the operation device **800** and the display device **900** are mounted is also referred to as a "console CON."

[0080] The constituent devices of the electric endoscope system **1000** will be described below.

[Insertion Manipulator **100**]

[0081] FIG. **3** is a diagram illustrating a distal end portion of an insertion section **110**.

[0082] The insertion manipulator **100** includes an insertion section **110**, an attachment portion **150**, a bending wire **160**, a built-in element **170**, and a scope **200** as illustrated in FIGS. **1** and **3**.

[0083] In the insertion manipulator **100**, an internal path (a lumen) **101** extending in a longitudinal direction (a longitudinal axis direction, an axial direction) A of the insertion manipulator **100** is formed from a distal end of the insertion section **110** to a proximal end of the attachment portion **150**. The bending wire **160** and the built-in element **170** are inserted into the internal path **101**.

[0084] The insertion section **110** is a thin longitudinal member that can be inserted into a lumen. The insertion section **110** includes a distal end portion **111**, a bending portion **112**, and a flexible portion **119**. The distal end portion **111**, the bending portion **112**, and the flexible portion **119** are sequentially connected from the distal end side A**1** to the proximal side A**2**. The insertion section **110** includes an outer sheath **118** which is disposed outermost.

[0085] FIG. **4** is a front view of the distal end portion **111** when seen from the distal end side A**1**.

[0086] The distal end portion **111** is formed in a cylindrical shape. The distal end portion **111** includes a first opening **111***a*, a second opening **111***b*, a water supply nozzle **111***d*, an air supply nozzle **111***e*, and a suction nozzle **111***f*. The first opening **111***a*, the second opening **111***b*, the water supply nozzle **111***d*, the air supply nozzle **111***e*, and the suction nozzle **111***f* are formed on a distal end surface of the distal end portion **111**.

[0087] FIG. **5** is a sectional view of the distal end portion **111**.

[0088] The built-in element **170** is inserted into the internal path **101**. The built-in element **170** includes a first channel tube **171**, a second channel tube **172**, an imaging cable **173**, a light guide **174**, a water supply tube **175**, an air supply tube **176**, and a suction tube **177**. In FIG. **5**, two treatment manipulators **400** that are inserted into the first channel tube **171** are also illustrated.

[0089] The first opening **111***a* is an opening that communicates with the first channel tube **171**. The first opening **111***a* is a circular opening in a front view when seen from the distal end side A**1**. The distal end of the treatment manipulators **400** inserted into the first channel tube **171** protrudes from and retract to the first opening **111***a*.

[0090] In the first opening **111***a*, a cutout portion **111***n* is formed at both ends in a direction (an LR direction which will be described later) perpendicular to a longitudinal direction A. As illustrated in FIG. **3**, the treatment manipulators **400** protruding from the first opening **111***a* to the distal end side A**1** can be inserted into the cutout portions **111***n*. The cutout portions **111***n* may not penetrate in the direction perpendicular to the longitudinal direction A as long as they are formed on the inner circumferential surface of the first opening **111***a*.

[0091] The second opening **111***b* is an opening that communicates with the second channel tube **172**. The second opening **111***b* is a circular opening in a front view when seen from the distal end side A**1**. A distal end of the treatment manipulator **400** inserted into the second channel tube **172** protrudes from and retracts to the second opening **111***b*. Devices inserted into the second channel tube **172** are not limited to the treatment manipulator **400** and include medical devices such as manual instruments.

[0092] An inner diameter D**1** of a part other than the cutout portions **111***n* in the first opening **111***a* is larger than an inner diameter D**2** of the second opening **111***b*. Specifically, the diameter D**1** of the part other than the cutout portions **111***n* in the first opening **111***a* is three to five times the inner diameter D**2** of the second opening **111***b*.

[0093] The water supply nozzle **111***d* is an opening that communicates with the water supply tube **175**. A liquid in a tank installed in the vicinity of the control device **700** is sent out from the water supply nozzle **111***d* via the water supply tube **175**.

[0094] The air supply nozzle **111***e* is an opening that communicates with the air supply tube **176**. A gas in a tank installed in the vicinity of the control device **700** is sent out from the air supply nozzle **111***e* via the air supply tube **176**.

[0095] The suction nozzle **111***f* is an opening that communicates with the suction tube **177**. A tank installed in the vicinity of the control device **700** sucks a gas or a liquid from the suction nozzle **111***f* via the suction tube **177**.

[Scope **200**]

[0096] The scope **200** is a unit that observes a lesion or the like and is attached to the distal end portion **111**. The scope **200** may be bendably attached to protrude from the distal end portion **111** to the distal end side A**1**. The scope **200** includes an imaging unit **201** and an illumination unit **202**.

[0097] The imaging unit **201** includes a stereo lens and an imaging element such as a CMOS and images an imaging subject. An imaging signal is sent to the video control device **600** via the imaging cable **173**.

[0098] The illumination unit **202** is connected to a light guide **174** that guides illumination light and emits illumination light illuminating an imaging subject.

[0099] In the electric endoscope system **1000**, the whole insertion manipulator **100** may be considered as an "endoscope." In addition, the scope **200**, the imaging cable **173**, and the light guide **174** may be considered as an "endoscope."

[0100] FIG. **6** is diagram illustrating the bending portion **112**.

[0101] The bending portion **112** includes a plurality of joint rings (also referred to as bending pieces) **115**, distal end portions **116** connected to the distal ends of the plurality of joint rings **115**, and an outer sheath **118**. The plurality of joint rings **115** are connected in the longitudinal direction A in the outer sheath **118**. The joint ring **115** at the distal end is connected to the distal end portion **111**. In the bending portion **112** illustrated in FIG. **3**, the outer sheath **118** is not illustrated.

[0102] FIG. **7** is a diagram illustrating joint rings **115**.

[0103] A joint ring **115** is a short cylindrical member formed of metal. A plurality of joint rings **115** are connected such that internal spaces of the neighboring joint rings **115** form a continuous space.

[0104] Each joint ring **115** includes a first joint ring **115***a* on the distal end side and a second joint ring **115***b* on the proximal side. The first joint ring **115***a* and the second joint ring **115***b* are connected by a first rotary pin **115***p* to be rotatable in a vertical direction (also referred to as an "UD direction") perpendicular to the longitudinal direction A.

[0105] In the neighboring joint rings **115**, the second joint ring **115***b* of the joint ring **115** on the distal end side and the first joint ring **115***a* of the joint ring **115** on the proximal side are connected by a second rotary pin **115***q* to be rotatable in a lateral direction (also referred to as an "LR direction") perpendicular to the longitudinal direction A and the UD direction.

[0106] The first joint rings **115***a* and the second joint rings **115***b* are alternately connected by the first rotary pin **115***p* and the second rotary pin **115***q*, and the bending portion **112** is bendable in a desired direction.

[0107] An upper wire guide **115***u* and a lower wire guide **115***d* are formed on the inner circumferential surface of the second joint ring **115***b*. The upper wire guide **115***u* and the lower wire guide **115***d* are disposed at both ends in the UD direction with the center axis O**1** in the longitudinal direction A interposed therebetween. A left wire guide **1151** and a right wire guide **115***r* are formed in the inner circumferential surface of the first joint ring **115***a*. The left wire guide **1151** and the right wire guide **115***r* are disposed at both ends in the LR direction with the center axis O**1** in the longitudinal direction A interposed therebetween.

[0108] The upper wire guide **115***u*, the lower wire guide **115***d*, the left wire guide **1151**, and the right wire guide **115***r* are formed such that through-holes into which the bending wire **160** is inserted are parallel to the longitudinal direction A.

[0109] The bending wire **160** is a wire for bending the bending portion **112**. The bending wire **160** extends to the attachment portion **150** via the internal path **101**. The bending wire **160** includes an upper bending wire **161***u*, a lower bending wire **161***d*, a left bending wire **1611** (see FIG. **3**), and a right bending wire **161***r*.

[0110] The upper bending wire **161***u* and the lower bending wire **161***d* are wires for bending the bending portion **112** in the UD direction. The upper bending wire **161***u* is inserted into the upper wire guide **115***u*. The lower bending wire **161***d* is inserted into the lower wire guide **115***d*.

[0111] The left bending wire **1611** and the right bending wire **161***r* are wires for bending the bending portion **112** in the LR direction. The left bending wire **1611** is inserted into the left wire guide **1151**. The right bending wire **161***r* is inserted into the right wire guide **115***r*.

[0112] The distal end of the bending wire **160** is fixed to the distal end portion **116** of the bending portion **112**. The bending portion **112** is bendable in a desired direction by attracting or relaxing the bending wires **160** (the upper bending wire **161***u*, the lower bending wire **161***d*, the left bending wire **1611**, and the right bending wire **161***r*).

[0113] The bending wire **160** and the built-in elements **170** (the first channel tube **171**, the second channel tube **172**, the imaging cable **173**, the light guide **174**, the water supply tube **175**, the air supply tube **176**, and the suction tube **177**) are inserted into the internal path **101** formed in the bending portion **112**. In FIG. **6**, built-in elements **170** other than the first channel tube **171** are not illustrated.

[0114] The flexible portion **119** is a tubular member which is longitudinal and flexible. The flexible portion **119** includes an outer sheath **118** which is disposed at the outermost. A distal end of the flexible portion **119** is connected to the bending portion **112**. The bending wire **160** and the built-in elements **170** (the first channel tube **171**, the second channel tube **172**, the imaging cable **173**, the light guide **174**, the water supply tube **175**, the air supply tube **176**, and the suction tube **177**) are inserted into the internal path **101** formed in the flexible portion **119**.

[0115] The attachment portion **150** is provided at a proximal end of the flexible portion **119** as illustrated in FIG. **1**. The attachment portion **150** is attached to the drive device **500** and the video control device **600**.

[First Channel Tube **171**]

[0116] As illustrated in FIG. **5**, the first channel tube **171** is a tube including a first treatment tool lumen **171***r* with a large diameter. The second channel tube **172** is a tube including a second treatment tool lumen **172***r*. The inner diameter D**1** of the first treatment tool lumen **171***r* is larger than the inner diameter D**2** of the second treatment tool lumen **172***r*. Specifically, the inner diameter D**1** of the first treatment tool lumen **171***r* is three to five time the inner diameter D**2** of the second treatment tool lumen **172***r*. As illustrated in FIG. **3**, two treatment manipulators **400** can be inserted into the first treatment tool lumen **171***r.*

[0117] The inner diameter D**12** of the first treatment tool lumen **171***r* is equal to or greater than ½ times the outer diameter of the outer sheath **118**. In an existing endoscope (for example, a nasal endoscope) in which the inner diameter of a treatment tool channel is relatively large, the outer diameter is about 6 mm and the inner diameter of the treatment tool channel is about 2.4 mm (about ⅖ times the outer diameter). In the insertion manipulator **100**, the inner diameter D**1** of the first treatment tool lumen **171***r* with respect to the outer diameter of the outer sheath **118** is larger than that in the existing endoscope. Accordingly, a large treatment manipulator **400** can be inserted into the first treatment tool lumen **171***r*, and a range of a manual operation is widened.

[0118] FIG. **8** is a diagram illustrating the first channel tube **171**. In FIG. **8**, the built-in elements **170** other than the first channel tube **171** are not illustrated.

[0119] The first channel tube **171** includes a proximal end channel tube **171**A that is disposed in the flexible portion **119** and a distal end channel tube **171**B that is disposed in the bending portion **112**. The proximal end channel tube **171**A and the distal end channel tube **171**B are connected to form the first treatment tool lumen **171***r.*

[0120] FIG. **9** is a sectional view of the proximal end channel tube **171**A.

[0121] The proximal end channel tube **171**A includes a coil sheath **171***a*, a blade tube **171***b*, a first fixed portion **171***c*, a second fixed portion **171***d*, and a regulating wire **171***e.*

[0122] The coil sheath **171***a* is a coil sheath that is flexibly bendable and has excellent kink-resistance. The coil sheath **171***a* forms the first treatment tool lumen **171***r* with a large diameter into which the treatment manipulator **400** is inserted.

[0123] The blade tube **171***b* is a tube in which metal strands, resin strands, or the like are woven in a blade shape and is disposed outside of the coil sheath **171***a*. It is preferable that the blade tube **171***b* be disposed to be in contact with an outer circumferential surface of the coil sheath **171***a.*

[0124] The proximal end channel tube **171**A is a tube with a double structure of the coil sheath **171***a* and the blade tube **171***b*. Accordingly, the proximal end channel tube **171**A also has excellent torque transmissivity based on the blade tube **171***b* while maintaining excellent kink-resistance based on the coil sheath **171***a*. By fitting the coil sheath **171***a* and the blade tube **171***b*, a frictional force between the coil sheath **171***a* and the blade tube **171***b* is further enhanced, and the torque transmissivity is further enhanced.

[0125] The coil sheath **171***a* is fixed to the blade tube **171***b* at a first fixed portion **171***c* on the distal end side A**1** in the axial direction A and a second fixed portion **171***d* on the proximal side A**2** with respect to the first fixed portion **171***c.*

[0126] The regulating wire **171***e* is connected to the first fixed portion **171***c* and the second fixed portion **171***d* and regulates a distance L**1** in the axial direction A between the first fixed portion **171***c* and the second fixed portion **171***d*. It is preferable that a plurality of regulating wires **171***e* be provided. The regulating wire **171***e* is a wire with high rigidity and is, for example, an NiTi wire.

[0127] Since the distance L**1** in the axial direction A between the first fixed portion **171***c* and the second fixed portion **171***d* is regulated by the regulating wire **171***e*, it is possible to prevent a decrease in torque transmissivity or a decrease in pushability due to expansion and contraction of the coil sheath **171***a* and the blade tube **171***b.*

[0128] The regulating wire **171***e* is provided to creep alternately inside and outside of the blade tube **171***b*. Accordingly, the regulating wire **171***e* can appropriately regulate the distance L**1** in the axial direction A between the first fixed portion **171***c* and the second fixed portion **171***d* regardless of a bent shape of the insertion section **110**.

[0129] The coil sheath **171***a* may be fixed to the blade tube **171***b* by the first fixed portion **171***c* and the second fixed portion **171***d* in a state in which the coil sheath is compressed in the axial direction A. In this case, the coil sheath **171***a* has a biasing force for separating the first fixed portion **171***c* and the second fixed portion **171***d* in the axial direction A. A tension which is an attraction in the axial direction A acts on the blade tube **171***b*. As a result, it is possible to prevent a decrease in torque transmissivity or a decrease in pushability due to contraction of the coil sheath **171***a* and the blade tube **171***b*. When this effect is not enough, the regulating wire **171***e* is not necessarily required.

[0130] As described above, the proximal end channel tube **171**A includes the first treatment tool lumen **171***r* with a large diameter, but has excellent kink-resistance, excellent torque transmissivity, and excellent pushability. Accordingly, even when the outer sheath **118** which is the outermost sheath of the flexible portion **119** is thin, the flexible portion **119** can maintain excellent kink-resistance, excellent torque transmissivity, and excellent pushability.

[0131] The distal end channel tube **171**B may have the same configuration as the proximal end channel tube **171**A. However, the bending portion **112** in which the distal end channel tube **171**B is provided includes the joint rings **115** with high rigidity and thus has excellent torque transmissivity and pushability. Accordingly, the distal end channel tube **171**B has only to include the coil sheath **171***a* forming the first treatment tool lumen **171***r* and may not include the blade tube **171***b* and the regulating wire **171***e*.

[0132] Since the outer sheath **118** can be formed of a thin film as described above, it is possible to achieve a decrease in diameter of the insertion manipulator **100**. However, when the first channel tube **171** does not have sufficient kink-resistance, torque transmissivity, or pushability, the outer sheath **118** may be a multi-layered tube including an outer coil sheath into which the first channel tube **171** is inserted and an outer blade tube which is disposed outside of the outer coil sheath.

[Drive Device **500**]

[0133] FIG. **10** is a functional block diagram illustrating the drive device **500**.

[0134] The drive device **500** includes an endoscope adapter **510**, an operation receiving unit **520**, an air supply/suction driving unit **530**, a drive unit **550**, and a drive controller **560**.

[0135] The endoscope adapter **510** is an adapter to which the insertion manipulator **100** is detachably connected. The endoscope adapter **510** connects the bending wire **160**, the suction tube **177**, the water supply tube **175**, and the air supply tube **176** to the drive device **500**.

[0136] The operation receiving unit **520** receives an operation input from the operation device **800** via the operation cable **801**. When the operation device **800** and the drive device **500** communicate in a wireless communication manner instead of a wired communication manner, the operation receiving unit **520** includes a known wireless receiver module.

[0137] The air supply/suction driving unit **530** is connected to the suction tube **177**, the water supply tube **175**, and the air supply tube **176** via the endoscope adapter **510**. The air supply/suction driving unit **530** includes a pump and supplies a liquid to the water supply tube **175**. The air supply/suction driving unit **530** supplies air to the air supply tube **176**. The air supply/suction driving unit **530** sucks air from the suction tube **177**.

[0138] The drive unit (actuator) **550** is connected to the bending wire **160** of the insertion manipulator **100** via the endoscope adapter **510**. The drive unit **550** includes a drive unit and an encoder which are not illustrated. The drive unit attracts or relaxes the bending wire **160** using a pulley or the like. The encoder detects an amount of attraction of the bending wire **160**. The detection result of the encoder is acquired by the drive controller **560** of the drive device **500**.

[0139] The drive unit (actuator) **550** may be connected to the treatment manipulator **400** to drive the treatment manipulator **400**.

[0140] The drive controller **560** controls the drive device **500** as a whole. The drive controller **560** acquires the operation input received by the operation receiving unit **520**. The drive controller **560** controls the air supply/suction driving unit **530** and the drive unit **550** on the basis of the acquired operation input.

[0141] The drive controller **560** is a program-executable computer including a processor **561**, a memory **562**, a storage unit **563** storing a program and data, and an input/output control unit **564**. The functions of the drive controller **560** are realized by causing the processor **561** to execute a program. At least some functions of the drive controller **560** may be realized by a dedicated logical circuit.

[0142] The drive controller **560** may further include an element in addition to the processor **561**, the memory **562**, the storage unit **563**, and the input/output control unit **564**. For example, the drive controller **560** may further include an image computing unit that performs some or all of image processing and image recognition processing. When the image computing unit is additionally provided, the drive controller **560** can perform specific image processing or image recognition processing at a high speed. The image computing unit may be mounted in a separate hardware device connected thereto via a communication line.

[Video Control Device **600**]

[0143] FIG. **11** is a functional block diagram illustrating the video control device **600**.

[0144] The video control device **600** includes an endoscope adapter **610**, an imaging processing unit **620**, a light source unit **630**, and a main controller **660**.

[0145] The endoscope adapter **610** is an adapter to which the insertion manipulator **100** is detachably connected. The endoscope adapter **610** connects the imaging cable **173** and the light guide **174** to the video control device **600**.

[0146] The imaging processing unit **620** is connected to the imaging cable **173**. The imaging processing unit **620** converts an imaging signal acquired from the imaging unit **201** of the scope **200** via the imaging cable **173** and the imaging cable **173** to a captured image.

[0147] The light source unit **630** is connected to the light guide **174**. The light source unit **630** generates illumination light which is applied to an imaging subject. The illumination light generated by the light source unit **630** is guided to an illumination unit **202** of the scope **200** via the light guide **174**.

[0148] The main controller **600** is a program-executable computer including a processor **661**, a memory **662**, a storage unit **663** that can store a program and data, and an input/output control unit **664**. The functions of the main controller **660** are realized by causing the processor **661** to execute a program. As least some functions of the main controller **660** may be realized by a dedicated logical circuit.

[0149] The main controller **660** can perform image processing on a captured image acquired by the imaging processing unit **620**. The main controller **660** can generate a GUI image or a CG image for providing information to an operator S. The main controller **660** can display the captured image, the GUI image, or the CG image on the display device **900**.

[0150] The main controller **660** is not limited to an integrated hardware device. For example, the main controller **660** may be constituted by separating a part thereof as a separate hardware device and connecting the separated hardware device thereto via a communication line. For example, the main controller **660** may be a cloud system that connects the separated storage units **663** via a communication line.

[0151] The main controller **660** may further include an element in addition to the processor **661**, the memory **662**, the storage unit **663**, and the input/output control unit **664**. For example, the main controller **660** may further include an image computing unit that performs some or all of image processing or image recognition processing which is performed by the processor **661**. Since the image computing unit is further provided, the main controller **660** can perform specific image processing or image recognition processing at a high speed. The image computing unit may be mounted in a separate hardware device connected thereto via a communication line.

[Operation of Electric Endoscope System **1000**]

[0152] Operations of the electric endoscope system **1000** according to the present embodiment will be described below. Specifically, manual operations of observing and treating a lesion formed in a wall of a large intestine using the electric endoscope system **1000** will be described.

[0153] An operator S inserts the insertion section **110** of the insertion manipulator **100** into the large intestine via an anal passage of a patient from its distal end. The operator S operates the operation device **800** while observing the captured image displayed on the display device **900** such that the insertion section **110** is moved to cause the distal end portion **111** to approach a lesion. The operator S operates the operation device **800** to bend the bending portion **112** according to necessity.

[0154] Since the insertion manipulator **100** has excellent kink-resistance, excellent torque transmissivity, and excellent pushability, the operator S can easily operate the insertion manipulator **100**.

[0155] The operator S inserts the treatment manipulator **400** into the first channel tube **171** or the second channel tube **172**. The operator S operates the operation device **800** while observing the captured image displayed on the display device **900** such that the treatment manipulator **400** is operated to treat the lesion.

[0156] Since the insertion manipulator **100** includes the first treatment tool lumen **171***r* with a large diameter, the operator S can appropriately perform treatment using the large treatment manipulator **400**.

[0157] After treating the lesion, the operator S removes the insertion manipulator **100** and the treatment manipulator **400** and ends the manual operation.

[0158] With the electric endoscope system **1000** according to the present embodiment, it is possible to more efficiently perform observation or treatment. Since the insertion manipulator **100** has excellent kink-resistance, excellent torque transmissivity, and excellent pushability, the operator S can easily operate the insertion manipulator **100**. Since the insertion manipulator **100** includes the first treatment tool lumen **171***r* with a large diameter, the operator S can appropriately perform treatment using the large treatment manipulator **400**.

[0159] While the first embodiment of the present invention has been described above in details with reference to the drawings, a specific configuration is not limited to this embodiment and includes a change in design without departing from the gist of the present invention. Elements described in the aforementioned embodiment and modified examples can be appropriately combined into a configuration.

Modified Example 1-1

[0160] FIG. **12** is a diagram illustrating a bending portion **112**A which is a modified example of the bending portion **112**. In FIG. **12**, the built-in elements **170** other than the first channel tube **171** are not illustrated.

[0161] The bending portion **112** does not include a plurality of joint rings **115** but includes a plurality of ring members **115**A. The plurality of ring members **115**A are arranged in the axial direction A. The built-in elements **170** including the first channel tube **171** are inserted into the plurality of ring members **115**A. The bending wire **160** is inserted into a wire guide **115***g* formed in the ring members **115**A.

[0162] FIG. **13** is a diagram illustrating the first channel tube **171** which is inserted into the bending portion **112**A.

[0163] The bending portion **112**A has lower torque transmissivity or lower pushability in comparison with the bending portion **112** in which the joint rings **115** are connected. Accordingly, a distal end portion of the first channel tube **171** into which the bending portion **112**A is inserted is preferably a tube with a double structure of the coil sheath **171***a* and the blade tube **171***b*. The distal end portion of the first channel tube **171** into which the bending portion **112**A is inserted preferably includes a regulating wire **171***e* for regulating a distance between the ends.

[0164] FIG. **14** is a diagram illustrating an insertion section **110**A which is a modified example of the insertion section **110**.

[0165] The insertion section **110**A includes an expanding and contracting portion **117** in addition to a distal end portion **111**, a bending portion **112**, and a flexible portion **119**. The expanding and contracting portion **117** is a member that connects the distal end portion **116** of the bending portion **112** to the distal end portion **111** and is driven by the drive unit (actuator) **550** to expand and contract in the longitudinal direction A. An outer circumferential portion of the expanding and contracting portion **117** is formed in a bellows shape. The operator S can accurately control a position of the distal end portion **111** in the longitudinal direction A with expansion and contraction of the expanding and contracting portion **117**.

[0166] FIG. **15** is a diagram illustrating a spiral tube **180**.

[0167] The operator may insert the insertion section **110** into a spiral tube **180** which is separate from the insertion section **110**. The spiral tube **180** is a tube in which a fin **181** is wound in a spiral shape on an outer circumference thereof. The spiral tube **180** is driven by the drive unit (actuator) **550** to be rotatable about a rotation axis extending in the longitudinal direction A. The operator can cause the spiral tube **180** and the insertion section **110** to move forward and rearward in the lumen by rotating the spiral tube **180**. For example, the operator can easily insert the distal end portion **111** of the insertion section **110** to a deep region of a large intestine.

[0168] FIG. **16** is a diagram illustrating a distal end portion **111**A which is a modified example of the distal end portion **111**.

[0169] The distal end portion **111**A includes a cutout portion **111***g* in which a part of a cylindrical body is cut out. The cutout portion **111***g* extends in the longitudinal direction A. A first opening **111***a* of the distal end portion **111**A is provided on the proximal side A**2** of the cutout portion **111***g*. The first opening **111***a* is open to the distal end side A**1**. The scope **200** is attached to the distal end of the distal end portion **111**A. The distal end portion **111**A includes a treatment camera **111***c* in addition to the scope **200** which is an insertion camera. The treatment camera **111***c* is provided on the distal end side A**1** of the cutout portion **111***g*. The first opening **111***a* and the treatment camera **111***c* are provided to face each other. The treatment camera **111***c* can image a situation in which the treatment manipulator **400** protruding from the first opening **111***a* teats a lesion. Since the direction which the treatment camera **111***c* faces and the direction in which the treatment manipulator **400** protrudes are opposite, the operator can perform treatment while normally viewing the distal end of the treatment manipulator **400**.

Second Embodiment

[0170] A hardness changing device **300** according to a second embodiment of the present disclosure will be described below with reference to FIGS. **17** to **25**. In the following description, the same constituents as in the above description will be referred to by the same reference signs, and repeated description thereof will be omitted.

[0171] FIG. **17** is a diagram a hardness changing device **300** which is inserted into the bending portion **112**.

[0172] The hardness changing device (rigidiser, introducer) **300** is a device including a hardness changing portion **310** into which the first treatment tool lumen **171***r* can be inserted. In FIG. **17**, the hardness changing device **300** is inserted into the first treatment tool lumen **171***r*. In FIG. **17**, the first channel tube **171** and the second channel tube **172** are not illustrated.

[0173] FIG. **18** is a diagram illustrating the entire configuration of the hardness changing device **300**.

[0174] The hardness changing device **300** includes a hardness changing portion **310**, an insertion section **320**, and a drive unit **340**.

[0175] FIGS. **19** and **20** are diagrams illustrating the hardness changing portion **310**.

[0176] The hardness changing portion **310** is a long member which can be inserted into the first treatment tool lumen **171***r* and in which hardness changes by performing a predetermined operation thereon. The hardness changing portion **310** includes a plurality of spinal portions **311** and a wire **312**. Each spinal portion **311** is formed in a bowl shape. The plurality of spinal portions **311** are connected in a longitudinal axis direction while overlapping each other. The wire **312** is inserted into the plurality of spinal portions **311** and is fixed to a spinal portion at the distal end. When the wire **312** is attracted to the proximal side as illustrated in FIG. **20**, neighboring spinal portions **311** come into close contact, and thus a frictional force is enhanced and the shape of the hardness changing portion **310** is fixed. By loosening the wire **312**, the shape of the hardness changing portion **310** changes.

[0177] The insertion section **320** is a long member that can be inserted into the first treatment tool lumen **171***r* and has flexibility. The insertion section **320** is provided on the proximal side of the hardness changing portion **310**.

[0178] The drive unit **340** is provided on the proximal side of the insertion section **320**. The drive unit **340** causes the insertion section **320** to move forward and rearward in the longitudinal axis direction and causes the insertion section **320** to rotate about the longitudinal axis. The drive unit **340** may be incorporated into the drive device **500**.

[0179] FIGS. **21** to **25** are diagrams illustrating operations of the hardness changing portion **310**.

[0180] As illustrated in FIG. **21**, the operator S inserts the distal end of the insertion section **110** of the insertion manipulator **100** into a large intestine of a patient via an anal passage. As illustrated in FIG. **22**, the operator S disposes the bending portion **112** in a part of the large intestine which is greatly bent. As illustrated in FIG. **23**, the operator S causes the hardness changing portion **310** with a variable shape to move forward and locates the hardness changing portion **310** in the bending portion **112**. The operator S fixes the shape of the hardness changing portion **310** inserted into the bending portion **112**. As illustrated in FIG. **24**, the operator S causes the insertion section **110** to move forward. The bending portion **112** moves forward along the hardness changing portion **310** with a fixed shape. The insertion manipulator **100** can smoothly pass through the part of the large intestine which is greatly bent. When the hardness changing portion **310** is not necessary, the operator S opens the wire of the hardness changing portion **310** and removes the hardness changing portion **310** from the bending portion **112** in a state in which it is softened as illustrated in FIG. **25**.

[0181] With the hardness changing device **300** according to the present embodiment, it is possible to more efficiently perform observation or treatment. The operator S can smoothly insert the insertion manipulator **100** to a target position using the hardness changing device **300**.

[0182] While the second embodiment of the present invention has been described above in details with reference to the drawings, a specific configuration is not limited to this embodiment and includes a change in design without departing from the gist of the present invention. Elements described in the aforementioned embodiment and modified examples can be appropriately combined into a configuration.

Modified Example 2-1

[0183] FIGS. **26** and **27** are diagrams illustrating a hardness changing portion **310**A which is a modified example of the hardness changing portion **310**. The hardness changing portion **310**A includes a tube **313** and a plurality of wire members **314** which are inserted into the tube **313**. Each wire member **314** is a metal strand or a resin strand. As illustrated in FIG. **27**, by making the internal space of the tube **313** have a negative pressure through suction and brining the plurality of wire members **314** into contact with each other, relative movement of the wire members **314** is curbed due to a frictional force between the plurality of wire members **314** and between the wire member **314** and the tube. Accordingly, the shape of the hardness changing portion **310**A is fixed. Since the frictional force is increased by roughening the surfaces of the wire members **314** to increase contact resistance, it is possible to further enhance hardness of the hardness changing portion **310**A with a fixed shape.

Modified Example 2-2

[0184] FIGS. **28** and **29** are diagrams illustrating a hardness changing portion **310**B which is a modified example of the hardness changing portion **310**. The hardness changing portion **310**B includes a plurality of spinal portions **311** and a tube **313**. The tube **313** is inserted into the plurality of spinal portions **311**. As illustrated in FIG. **29**, by making the internal space of the tube **313** have a positive pressure through suction such that the tube **313** expands, a frictional force between the neighboring spinal portions **311** is increased, and the shape of the hardness changing portion **310**B is fixed. Since the hardness changing portion **310**B makes the internal space of the tube **313** have a positive pressure, it is possible to easily increase the frictional force between the plurality of spinal portions **311** and to more strength the hardness changing portion **310**B in comparison with a mode in which the internal space of the tube **313** is made to have a negative pressure. In the mode in which the internal space of the tube **313** is made to have a negative pressure, the pressure has only to be reduced by only the atmospheric pressure. However, when the internal space of the tube **313** is made to have a positive pressure, such limitation is removed, and thus it is possible to apply a larger pressure.

Modified Example 2-3

[0185] FIGS. **30** and **31** are diagrams illustrating a hardness changing portion **310**C which is a modified example of the hardness changing portion **310**. The hardness changing portion **310**C includes an outer tube **315**, an inner tube **316**, and a plurality of cables **317**. The inner tube **316** is inserted into the inner space of the outer tube **315**. The plurality of cables **317** are inserted into a space V interposed between the outer tube **315** and the inner tube **316**. As illustrated in FIG. **31**, by making the space V have a negative pressure through suction, the plurality of cables **317** come into contact with each other, and the shape of the hardness changing portion **310**C is fixed. The space V may be made to have a negative pressure using a hydraulic pressure.

Modified Example 2-4

[0186] FIG. **32** is a diagram illustrating a hardness changing portion **310**D which is a modified example of the hardness changing portion **310**. The hardness changing portion **310**D includes a plurality of spinal portions **311** and a shape memory alloy wire **318**. In the hardness changing portion **310**D, when the shape memory alloy wire **318** contracts with supply of electric power, the spinal portions **311** come into close contact with each other similarly to the hardness changing portion **310**, and thus the hardness changing portion **310**D can switch between a state in which the shape is fixed and a state in which the shape is variable.

Third Embodiment

[0187] A treatment manipulator **400** according to a third embodiment of the present disclosure will be described below with reference to FIGS. **33** to **40**. In the following description, the same constituents as in the above description will be referred to by the same reference signs, and repeated description thereof will be omitted.

[0188] FIG. **33** is a diagram illustrating a high-frequency knife **430**.

[0189] The treatment manipulator **400** includes a bendable treatment tool arm **410** and treatment tools (a forceps **420**, a high-frequency knife **430**, a local injection needle **440**, and a basket **450**) which are inserted into the treatment tool arm **410**.

[0190] The treatment tool arm **410** is a hollow long member. Similarly to the bending portion **112** of the insertion manipulator **100**, the treatment tool arm **410** includes a plurality of joint rings (also referred to as bending pieces) **415** and is driven with a wire or the like to be bendable in the vertical direction and the lateral direction.

[0191] The treatment tools (such as the forceps **420**, the high-frequency knife **430**, the local injection needle **440**, and the basket **450**) can be inserted into the internal space (a lumen or a channel) of the treatment tool arm **410**. The treatment tools are manually bendable, but may not have an active bending function.

[0192] FIG. **34** is a diagram illustrating the high-frequency knife **430** for putting a marking M.

[0193] An operator S inserts the high-frequency knife **430** into the treatment tool arm **410** and causes the distal end of the high-frequency knife **430** to protrude from the distal end of the treatment tool arm **410**. The operator S locates the distal end of the high-frequency knife **430** at a desired position by bending the treatment tool arm **410**. The operator S cauterize biological tissue with the distal end portion of the high-frequency knife **430** and puts a marking M on the biological tissue.

[0194] Since the first treatment tool lumen **171***r* has a large diameter, two treatment tool arms **410** can be inserted thereinto. For example, as illustrated in FIG. **34**, the operator S can insert the high-

frequency knife **430** into one treatment tool arm **410**, insert the forceps **420** into the other treatment tool arm **410**, and treat a target part using two treatment tools.
[0195] FIG. **35** is a diagram illustrating a local injection needle **440**.
[0196] The operator S removes the high-frequency knife **430** from the treatment tool arm **410**. Then, the operator S inserts the local injection needle **440** into the treatment tool arm **410** and causes the distal end of the local injection needle **440** to protrude from the distal end of the treatment tool arm **410**.
[0197] FIG. **36** is a diagram illustrating the local injection needle **440** which locally injects a liquid.
[0198] The operator S bends the treatment tool arm **410** and locates the distal end of the local injection needle **440** at a desired position. The operator S pierces the biological tissue with the distal end of the local injection needle **440** and locally injects a liquid.
[0199] FIGS. **37** and **38** are diagrams illustrating a high-frequency knife **430** for cutting.
[0200] The operator S removes the local injection needle **440** from the treatment tool arm **410**. Then, the operator S inserts the high-frequency knife **430** into the treatment tool arm **410**. The operator S attracts biological tissue using the forceps **420**. The operator S cuts a target part of the biological tissue with the high-frequency knife **430** while checking the marking M.
[0201] FIG. **39** is a diagram illustrating a basket **450**.
[0202] The operator S removes the two treatment tool arms **410** along with the treatment tools. Then, the operator S inserts a large-diameter treatment tool arm **410**A into the first treatment tool lumen **171***r*. The operator S inserts the basket **450** into the large-diameter treatment tool arm **410**A. The basket **450** has a larger outer diameter than that of the high-frequency knife **430** or the local injection needle **440**. The large-diameter treatment tool arm **410**A has a larger inner diameter of a hollow portion than that of the treatment tool arm **410** and enables the basket **450** to be inserted thereinto. Only a large basket **450** may be directly inserted into the first treatment tool lumen **171***r* without using the large-diameter treatment tool arm **410**A.
[0203] FIG. **40** is a diagram illustrating the basket **450** for recovering a target part T.
[0204] The operator S recovers a target part T which has been cut and removed using the basket **450**. The operator S can efficiently recover the target part T using the large basket **450**. Since the first treatment tool lumen **171***r* has a large diameter, the large basket **450** larger than a basket used in the related art can be inserted thereinto. Since the first treatment tool lumen **171***r* has a large diameter, a larger amount of target part T or tissue than that in the related art can be easily recovered at a time without being caught by the first treatment tool lumen **171***r*. Since the first treatment tool lumen **171***r* has a large diameter, larger treatment tools than those in the related art in addition to the basket can be inserted thereinto.
[0205] With the treatment manipulator **400** according to the present embodiment, it is possible to more efficiently perform observation or treatment. Since the insertion manipulator **100** has the first treatment tool lumen **171***r* with a large diameter, two treatment manipulators **400** can be simultaneously inserted and used. Since the treatment manipulator **400** with a large diameter can be inserted into the first treatment tool lumen **171***r*, a multi-functional and high-performance large treatment manipulator **400** can be used.
[0206] While the third embodiment of the present invention has been described above in details with reference to the drawings, a specific configuration is not limited to this embodiment and includes a change in design without departing from the gist of the present invention. Elements described in the aforementioned embodiment and modified examples can be appropriately combined into a configuration.
Fourth Embodiment
[0207] A treatment manipulator **400**B according to a fourth embodiment of the present disclosure will be described below with reference to FIGS. **41** to **42**. In the following description, the same constituents as in the above description will be referred to by the same reference signs, and repeated description thereof will be omitted.

[0208] FIG. **41** is a diagram illustrating a treatment manipulator **400**B.
[0209] Similarly to the treatment manipulator **400** according to the aforementioned embodiment, the treatment manipulator **400**B is a device that is inserted into the first channel tube **171** or the like of the insertion manipulator **100** and is inserted into a lumen of a patient to treat a lesion. The treatment manipulator **400**B is driven by the drive unit (actuator) **550** or the like. The treatment manipulator **400**B is different from the treatment manipulator **400** in that two treatment tool arms **410** are arranged on a distal end surface of one manipulator flexible portion **417**. The treatment manipulator **400**B can be more easily inserted in comparison with the treatment manipulator **400**.
[0210] The treatment manipulator **400**B includes a treatment tool (such as a forceps **420**, a high-frequency knife **430**, a local injection needle **440**, or a basket **450**), a treatment tool arm **410**, an artificial muscle **470**, and a wire **480**.
[0211] The treatment tool arm **410** is a hollow long member. The treatment tool arm **410** includes a first bending portion **411** provided on the distal end side A**1** and a second bending portion **412** provided on the proximal side A**2**. In the use state of the treatment manipulator **400**B illustrated in FIG. **41**, the first bending portion **411** is located on the distal end side A**1** with respect to the distal end surface of the distal end portion **111**. A part of the second bending portion **412** protrudes from the distal end portion **111** to the distal end side A**1**.
[0212] The first bending portion **411** includes a plurality of joint rings (also referred to as wrist joints) **415** similarly to the bending portion **112** of the insertion manipulator **100** and can be driven by the artificial muscle **470** such that it is bendable in the vertical direction and the lateral direction.
[0213] As illustrated in FIG. **41**, the second bending portion **412** includes a shoulder joint **416** that greatly bends the treatment tool arm **410** in the lateral direction (LR direction). The shoulder joint **416** bends the second bending portion **412** in an S-shape when seen in the vertical direction (UD direction). As illustrated in FIG. **41**, the distal ends of the second bending portions **412** of two treatment tool arms **410** are bent and separated from each other in the lateral direction. Here, a distance between the center axes at the distal ends of the second bending portions **412** of the two treatment tool arms **410** is defined as a “distance (shoulder width) L.”
[0214] When a target part is treated using two treatment tool arms **410**, the operator S can secure space enough for treatment and easily treat the target part by appropriately separating the positions of the proximal ends of the two first bending portions **411**. Therefore, the operator S greatly bends the second bending portions **412** such that the distance L is appropriately increased. At least a part of the two treatment manipulators **400** can be inserted into cutout portions **111***n* and spread outward in the lateral direction. Accordingly, the distance L may be larger than the outer diameter of the distal end portion **111** of the insertion manipulator **100**.
[0215] The artificial muscle **470** constitutes at least a part of a mechanism for bending a treatment tool. The artificial muscle **470** is, for example, a Mckibben artificial muscle. The artificial muscle **470** is attached to the first bending portion **411** and bends the first bending portion **411**. In order to bend the first bending portion **411** in the vertical direction and the lateral direction, a plurality of artificial muscles **470** are attached to the first bending portion **411**.
[0216] In the use state of the treatment manipulator **400**B illustrated in FIG. **41**, the first bending portion **411** is disposed on the distal end side A**1** with respect to the distal end surface of the distal end portion **111**. Accordingly, the artificial muscle **470** is also disposed on the distal end side A**1** with respect to the distal end surface of the distal end portion **111**.
[0217] FIG. **42** is a sectional view of the treatment manipulator **400**B.
[0218] A tube **471** for supplying a fluid for operating the artificial muscle **470** is attached to the artificial muscle **470**. The artificial muscle **470** is contracted with the fluid supplied from the tube **471**.
[0219] The wire **480** is attached to the second bending portion **412**. By attracting the second bending portion **412** from the proximal side A**2**, the second bending portion **412** is bent.
[0220] The first bending portion **411** is bendably driven by the artificial muscle **470**. The artificial

muscle **470** can more precisely bend the first bending portion **411** in comparison with the wire **480**. Accordingly, the artificial muscle **470** is suitable for bendable driving of the first bending portion **411** in which precise operations for treatment are necessary. The artificial muscle **470** is attached to the first bending portion **411** and directly bends the first bending portion **411**. Accordingly, it is possible to accurately bend the first bending portion **411** without being affected by the shape on the proximal side of the first bending portion **411**.

[0221] The second bending portion **412** is bendably driven by the wire **480**. The second bending portion **412** does not require a very accurate operation as long as it can greatly bend the shoulder joint **416**. Accordingly, it is possible to satisfactorily bend the second bending portion **412** even using the wire **480**. Through driving using the wire **480**, it is possible to easily secure a longer stroke in comparison with the artificial muscle **470**. Accordingly, the wire **480** is suitable for bendable driving of the second bending portion **412** in which a great bending operation is necessary.

[0222] With the treatment manipulator **400**B according to the present embodiment, it is possible to more efficiently perform observation or treatment. Since the first bending portion **411** which is bendably driven by the artificial muscle **470** can precisely bend the treatment tool, the operator S can more efficiently perform observation or treatment.

[0223] While the fourth embodiment of the present invention has been described above in details with reference to the drawings, a specific configuration is not limited to this embodiment and includes a change in design without departing from the gist of the present invention. Elements described in the aforementioned embodiment and modified examples can be appropriately combined into a configuration.

Modified Example 4-1

[0224] FIG. **43** is a diagram illustrating a treatment manipulator **400**C which is a modified example of the treatment manipulator **400**B. The treatment manipulator **400**C includes a treatment tool and an artificial muscle **470** and does not include a treatment tool arm **410**. The artificial muscle **470** is attached to the treatment tool using a ring member **472**. A member which is bent with a compression force, for example, a tube or a coil, can be employed instead of the plurality of joint rings **415** illustrated in FIG. **41**, and the artificial muscle **470** can be mounted around the member. Instead of a configuration in which the artificial muscle **470** is disposed outside of the joint rings **415**, a member which is bent with a compression force, for example, a tube or a coil, may be disposed outside of the artificial muscle **470**.

Modified Example 4-2

[0225] FIG. **44** is a diagram illustrating an artificial muscle **470** which is disposed at a different position.

[0226] The artificial muscle **470** may be provided on the proximal side A**1** with respect to the second bending portion **412**. The first bending portion **411** and the artificial muscle **470** are connected by a distal end wire **473**. The artificial muscle **470** bends the first bending portion **411** by causing the distal end wire **473** to move forward and rearward. The artificial muscle **470** illustrated in FIG. **44** is disposed in the distal end portion **111** on the distal end side A**1** with respect to the bending portion **112**. The artificial muscle **470** is disposed in the distal end portion **111** which is not bent. Accordingly, even when the bending portion **120** is bent, the artificial muscle **470** is not bent.

Modified Example 4-3

[0227] FIG. **45** is a diagram illustrating an artificial muscle **470** which is disposed at a different position.

[0228] The artificial muscle **470** illustrated in FIG. **45** is disposed in the bending portion **112**. The artificial muscle **470** is disposed in a non-bending area E which is not bent and which is interposed between a first rotary pin **115***p* and a second rotary pin **115***q*. Accordingly, even when the bending portion **120** is bent, the artificial muscle **470** is not bent.

Modified Example 4-4

[0229] FIG. **46** is a diagram illustrating an artificial muscle **470** which is disposed at a different position.

[0230] The artificial muscle **470** illustrated in FIG. **46** is disposed in a flexible portion **119** on the proximal side A**2** with respect to the bending portion **112**. The artificial muscle **470** is disposed in the flexible portion **119** which is not greatly bent. Accordingly, even when the bending portion **120** is bent, the artificial muscle **470** is not bent.

Modified Example 4-5

[0231] FIG. **47** is a diagram illustrating an artificial muscle **470** which is disposed at a different position.

[0232] The artificial muscle **470** illustrated in FIG. **47** is disposed in the flexible portion **119** on the proximal side A**2** with respect to the bending portion **112** and is located at a position separated by a distance L**3** from the proximal end with respect to the bending portion **112**. The distance L**3** is longer than a moving distance of the treatment manipulator **400**B at the time of treatment. Accordingly, even when the treatment manipulator **400**B is intended to move forward and rearward at the time of treatment, the artificial muscle **470** is not inserted into the bending portion **112**. Accordingly, even when the bending portion **120** is bent, the artificial muscle **470** is not bent.

Modified Example 4-6

[0233] FIG. **48** is a diagram illustrating an artificial muscle **470** which is disposed at a different position.

[0234] As illustrated in FIG. **48**, a plurality of artificial muscles **470** may be arranged in the axial direction A. It is preferable that some artificial muscles **470** be disposed in a non-bending area. It is not necessary to arrange the plurality of artificial muscles **470** in the radial direction and it is possible to decrease the diameter of the treatment manipulator **400**B.

Modified Example 4-7

[0235] FIG. **49** is a diagram illustrating a treatment manipulator **400**D which is a modified example of the treatment manipulator **400**B. The treatment manipulator **400**D does not include a wire **480** and bends all bending portions using the artificial muscle **470**. The operator S can precisely bend all the bending portions.

Fifth Embodiment

[0236] A suture device **400**E according to a fifth embodiment of the present disclosure will be described below with reference to FIGS. **50** to **60**. In the following description, the same constituents as in the above description will be referred to by the same reference signs, and repeated description thereof will be omitted.

[0237] FIG. **50** is a diagram illustrating a suture device **400**E that protrudes from the distal end portion **111** of the insertion manipulator **100**. The suture device **400**E is a device that is inserted into the first channel tube **171** or the like of the insertion manipulator **100** and sutures a defective part D in biological tissue using a thread TH similarly to the treatment manipulator **400** according to the aforementioned embodiment. The suture device **400**E has a larger outer diameter than the treatment manipulator **400** and thus is inserted into the first treatment tool lumen **171***r* instead of the second treatment tool lumen **172***r*. The suture device **400**E is driven by the drive unit (actuator) **550** or the like.

[0238] In FIG. **50**, the scope **200** is disposed at a position protruding from the distal end portion **111**. The scope **200** can also be disposed at a position which is accommodated in the distal end portion **111**. That is, the scope **200** can move forward and rearward with respect to the insertion manipulator **100** independently from the suture device **400**E. As illustrated in FIG. **50**, the scope **200** can be bent toward the center axis of the first channel tube (lumen) **171** in a state in which it has protruded from the distal end portion **111**.

[0239] The suture device **400**E includes a hollow bendable arm **410**E, a distal end portion **411**E that is provided at the distal end of the arm **410**E, a first jaw **412**E, a second jaw **413**E, and a needle **414**E.

[0240] The arm **410**E includes a bending mechanism such as an artificial muscle **470** or a wire **480** similarly to the treatment tool arm **410** of the treatment manipulator **400**. The arm **410**E includes a hollow portion with a large diameter similarly to the large-diameter treatment tool arm **410**A.
[0241] The distal end portion **411**E is formed in a substantially cylindrical shape. An internal space **411***s* of the distal end portion **411**E communicates with the hollow portion of the arm **410**E. The internal space **411***s* of the distal end portion **411**E and the hollow portion of the arm **410**E form a channel **420**E into which a treatment device such as an anchor applier **400**F and a thread attraction tool **400**G which will be described later can be inserted. The distal end portion **411**E includes a cutout **411***n* communicating with the internal space **411***s* on the upper side. A device which is inserted into the channel **420**E is not limited thereto, and various treatment tools including a forceps for attracting tissue and applying an appropriate tension thereto, a treatment tool with a screw-shaped (helical) distal end, and a local injection needle for locally injecting a liquid can be inserted thereinto.
[0242] The first jaw (an upper jaw portion) **412**E and the second jaw (a lower jaw portion) **413**E are provided in the distal end portion **411**E and can be opened and closed in the vertical direction. In the present embodiment, the first jaw **412**E is rotatably attached to the distal end portion **411**E, and the second jaw **413**E is non-rotatably attached to the distal end portion **411**E. A rotary axis RO of the first jaw **412**E extends in the lateral direction. The first jaw **412**E and the second jaw **413**E deliver the needle **414**E.
[0243] The first jaw **412**E includes a jaw body **412***b* which is formed in a substantially U-shape and a rotary arm **412***a* which extends in an opening and closing direction P. The needle **414**E is detachably attached (latched) to the distal end of the rotary arm **412***a*.
[0244] FIG. **51** is a diagram illustrating a captured image which is captured by the scope **200**.
[0245] A space surrounded by sides of the jaw body **412***b* formed in a substantially U-shape communicates with a space surrounded by the cutout **411***n* of the distal end portion **411**E to form a viewing field space (a penetration space) VS penetrating the first jaw **412**E in the opening and closing direction P. The imaging unit **201** of the scope **200** can image a suturing part or the needle **414**E through the viewing field space VS. Even when the first jaw **412**E rotates in the opening and closing direction P, the first jaw **412**E includes the viewing field space VS and thus does not interfere with the viewing field of the imaging unit **201**.
[0246] The second jaw (lower jaw portion) **413**E is accommodated in the distal end portion **411**E. The second jaw **413**E may be provided to protrude from the distal end portion **411**E to the distal end side A**1**. The second jaw **413**E includes a needle receiving portion **413***a* and a needle locking mechanism **413***b*.
[0247] The needle receiving portion **413***a* is a portion that receives the needle **414**E attached (latched) to the distal end of the rotary arm **412***a* of the first jaw **412**E.
[0248] The needle locking mechanism **413***b* is a mechanism that locks the needle **414**E received in the needle receiving portion **413***a*. The needle locking mechanism **413***b* is driven by the drive unit (actuator) **550** or the like.
[0249] FIGS. **52** to **55** are diagrams illustrating the needle **414**E which is delivered from the first jaw **412**E to the second jaw **413**E. As illustrated in FIG. **52**, the rotary arm **412***a* of the first jaw **412**E is closed downward. As illustrated in FIG. **53**, the needle **414**E is received in the needle receiving portion **413***a*. As illustrated in FIG. **54**, the needle locking mechanism **413***b* locks the needle **414**E. At this time, the needle locking mechanism **413***b* releases connection between the rotary arm **412***a* and the needle **414**E. As illustrated in FIG. **55**, the rotary arm **412***a* of the first jaw **412**E is opened upward. The needle **414**E departs from the rotary arm **412***a* of the first jaw **412**E and is located in the needle receiving portion **413***a*.
[0250] FIG. **56** is a diagram illustrating a sutured defective part D.
[0251] The operator S pierces a circumferential edge of the defective part D with the rotary arm **412***a* to which the needle **414**E is attached (latched) and puts the thread TH on the circumferential

edge of the defective part D. As illustrated in FIG. **56**, the operator S inserts the anchor applier **400**F into the channel **420**E after putting the thread TH on the circumferential edge of the defective part D.
[0252] The anchor applier **400**F can be inserted into the channel **420**E of the suture device **400**E and protrude from the channel **420**E to the distal end side A**1**. The anchor applier **400**F includes a long body **410**F, a first anchor **411**F, a second anchor **412**F, a plug **413**F, and a blade **414**F (see FIG. **59**).
[0253] The first anchor **411**F and the second anchor **412**F are detachably attached to the distal end of the body **410**F. The first anchor **411**F and the second anchor **412**F are provided in parallel in the vertical direction. The first anchor **411**F and the second anchor **412**F have the same shape. The first anchor **411**F (and the second anchor **412**F) is formed in a substantially U-shape, and a distal end space (a needle insertion space) S**1** into which the needle **414**E can be inserted and a proximal space (a thread insertion space) S**2** into which a thread TH can be inserted are formed therein. The distal end space S**1** and the proximal space S**2** communicate with each other.
[0254] The plug **413**F is detachably provided in the proximal space S**2**. When the plug **413**F is disposed in the proximal space S**2**, the first anchor **411**F and the second anchor **412**F are pressed outward by the plug **413**F, and the proximal space S**2** is widened. At this time, the first anchor **411**F and the second anchor **412**F are not fixed to the thread TH.
[0255] The blade **414**F is provided to move forward and rearward on the distal end side A**1** of the body **410**F. The blade **414**F is provided to move forward and rearward between the first anchor **411**F and the second anchor **412**F. The blade **414**F includes an edge **414***a* (see FIG. **59**) and can cut the thread TH.
[0256] FIGS. **57** to **60** are diagrams illustrating operations of the anchor applier **400**F.
[0257] As illustrated in FIG. **56**, the operator S causes the needle **414**E to pass through the distal end space S**1** of the anchor applier **400**F protruding from the channel **420**E to the distal end side A**1**. The thread TH is inserted into the distal end space S**1**.
[0258] As illustrated in FIG. **57**, the operator S inserts the thread attraction tool **400**G into the channel **420**E. The anchor applier **400**F and the thread attraction tool **400**G can be simultaneously inserted into the channel **420**E. For example, the thread attraction tool **400**G includes a hooking portion with a hook shape or a helical coil shape at its distal end. The operator S attracts the thread TH to the proximal side A**2** using the thread attraction tool **400**G and moves the thread TH from the distal end space S**1** to the proximal space S**2**.
[0259] As illustrated in FIG. **58**, the operator S attracts the plug **413**F to the proximal side A**2** and pulls the plug **413**F from the first anchor **411**F and the second anchor **412**F. The proximal space S**2** of the first anchor **411**F is narrowed and puts the thread TH therebetween. The first anchor **411**F is fixed to a first fixed portion of the thread TH which is inserted into the proximal space S**2**. The proximal space S**2** of the second anchor **412**F is narrowed and puts the thread TH therebetween. The second anchor **412**F is fixed to a second fixed portion of the thread TH which is inserted into the proximal space S**2**.
[0260] As illustrated in FIG. **59**, the operator S attracts the blade **414**F to the proximal side A**2** and cuts the thread TH. The blade **414**F cuts the thread TH between the first fixed portion and the second fixed portion. The thread TH on the defective part D side which is one end of the cut thread TH is referred to as a first thread TH**1**. The thread TH on the needle **414**E side which is the other end of the cut thread TH is referred to as a second thread TH**2**. The first anchor **411**F is attached to the first thread TH**1**. The second anchor **412**F is attached to the second thread TH**2**. The operator S removes the anchor applier **400**F and the thread attraction tool **400**G from the channel **420**E.
[0261] As illustrated in FIG. **60**, the first anchor **411**F attached to the first thread TH**1** serves as a thread stopper of ligation (termination) of the first thread TH**1** suturing the defective part D. The second anchor **412**F attached to the second thread TH**2** serves as a thread stopper of a start end of next suturing. The operator S can adjust a length of the second thread TH**2** which is used for next

suturing by adjusting an amount of attraction of the thread TH using the thread attraction tool **400**G and adjusting a length the thread TH between the first fixed portion and the second fixed portion.
[0262] With the suture device **400**E according to the present embodiment, it is possible to more efficiently perform observation or treatment. The operator S can successively perform installation of the first anchor **411**F serving as an end of suturing, attraction of the thread TH, and installation of the second anchor **412**F serving as a start of next suturing in a lumen. Accordingly, it is not necessary to perform preparation of thread stoppage or a complex operation using a knot pusher and it is possible to easily perform suturing to shorten a suturing time.
[0263] While the fifth embodiment of the present invention has been described above in details with reference to the drawings, a specific configuration is not limited to this embodiment and includes a change in design without departing from the gist of the present invention. Elements described in the aforementioned embodiment and modified examples can be appropriately combined into a configuration.
[0264] Some or all of the aforementioned functions may be realized by recording a program in each embodiment on a computer-readable recording medium and causing a computer system to read and execute the program recorded on the recording medium. The “computer system” includes an operating system (OS) or hardware such as peripherals. The “computer-readable recording medium” is a portable medium such as a flexible disk, a magneto-optical disc, a ROM, or a CD-ROM or a storage device such as a hard disk incorporated into a computer system. The “computer-readable recording medium” may include a medium that dynamically holds a program for a short time such as a communication line when the program is transmitted via a network such as the Internet or a communication circuit line such as a telephone line and a medium that holds a program for a predetermined time such as a volatile memory in a computer system serving as a server or a client in that case. The program may be a program for realizing some of the aforementioned functions. The program may be a program for realizing the aforementioned functions in combination with another program stored in advance in the computer system.
[0265] The present invention can be applied to a medical system that performs observation and treatment in a hollow organ.

Claims

1. A medical manipulator comprising: an outer sheath; and a channel tube that is inserted into the outer sheath, wherein at least a part of the channel tube includes a coil sheath forming a lumen into which a treatment manipulator or a treatment tool is inserted, and a blade tube disposed outside of the coil sheath.
2. The medical manipulator according to claim 1, wherein the coil sheath is fixed to the blade tube at a first fixed portion on a distal end side in an axial direction and a second fixed portion on a proximal side with respect to the first fixed portion, and wherein the channel tube includes a wire for regulating a distance in the axial direction between the first fixed portion and the second fixed portion.
3. The medical manipulator according to claim 1, wherein the coil sheath is fixed to the blade tube at a first fixed portion on a distal end side in an axial direction and a second fixed portion on a proximal side with respect to the first fixed portion, and wherein the coil sheath has a biasing force for separating the first fixed portion and the second fixed portion in the axial direction.
4. The medical manipulator according to claim 1, wherein an inner diameter of the lumen is equal to or greater than ½ times an outer diameter of the outer sheath.
5. The medical manipulator according to claim 1, further comprising a second channel tube that is inserted into the outer sheath, wherein the second channel tube includes a second lumen into which the treatment manipulator or the treatment tool is inserted, and wherein an inner diameter of the lumen is three times to five times an inner diameter of the second lumen.

6. The medical manipulator according to claim 1, further comprising: a bendable bending portion that is disposed at a distal end portion of the outer sheath; and a bending wire that is connected to the bending portion.
7. The medical manipulator according to claim 1, wherein the outer sheath is formed of a thin film.
8. The medical manipulator according to claim 1, wherein the outer sheath includes: an outer coil sheath into which the channel tube is inserted; and an outer blade tube that is disposed outside of the outer coil sheath.
9. The medical manipulator according to claim 1, further comprising a scope that is provided in a distal end portion of the outer sheath and includes an imaging unit.
10. The medical manipulator according to claim 9, wherein the scope is able to protrude from and retract to the distal end portion.
11. The medical manipulator according to claim 10, wherein the scope is bendable toward an axis of the channel tube in a state in which the scope has protruded from the distal end portion.
12. A medical manipulator system comprising: a medical manipulator; and a control device including a drive device that drives the medical manipulator, wherein the medical manipulator includes an outer sheath, and a channel tube that is inserted into the outer sheath, wherein at least a part of the channel tube includes a coil sheath forming a lumen into which a treatment manipulator or a treatment tool is inserted, and a blade tube disposed outside of the coil sheath.
13. The medical manipulator system according to claim 12, wherein the coil sheath is fixed to the blade tube at a first fixed portion on a distal end side in an axial direction and a second fixed portion on a proximal side with respect to the first fixed portion, and wherein the channel tube includes a wire for regulating a distance in the axial direction between the first fixed portion and the second fixed portion.
14. The medical manipulator system according to claim 12, wherein the coil sheath is fixed to the blade tube at a first fixed portion on a distal end side in an axial direction and a second fixed portion on a proximal side with respect to the first fixed portion, and wherein the coil sheath has a biasing force for separating the first fixed portion and the second fixed portion in the axial direction.
15. The medical manipulator system according to claim 12, wherein an inner diameter of the lumen is equal to or greater than ½ times an outer diameter of the outer sheath.
16. The medical manipulator system according to claim 12, further comprising a second channel tube that is inserted into the outer sheath, wherein the second channel tube includes a second lumen into which the treatment manipulator or the treatment tool is inserted, and wherein an inner diameter of the lumen is three times to five times an inner diameter of the second lumen.
17. The medical manipulator system according to claim 12, further comprising: a bendable bending portion that is disposed at a distal end portion of the outer sheath; and a bending wire that is connected to the bending portion, wherein the drive device drives the bending wire to bend the bending portion.
18. The medical manipulator system according to claim 12, wherein the outer sheath is formed of a thin film.
19. The medical manipulator system according to claim 12, wherein the outer sheath includes: an outer coil sheath into which the channel tube is inserted; and an outer blade tube that is disposed outside of the outer coil sheath.
20. The medical manipulator system according to claim 12, further comprising a scope that is provided in a distal end portion of the outer sheath and includes an imaging unit, wherein the drive device drives the scope to protrude from and retract to the distal end portion.